

Program ω Plan: The State

Draft for hostile review — frozen upon the reviewer’s confirmation. No stage computation runs before the review completes. Companion to papers I-V (closed), the synthesis (closed), and the registry. All standing disciplines carry over unchanged: two implementations for every load-bearing number, pre-registration before computation, tag discipline in prose, the diff protocol for documentary work, instruments defined before values are seen (P6), reproduce-before-explain, the falsifier-visibility check, and the error ledger as part of every record.

0. Charter

The chirality arc ended by handing everything undecided to one address: the state. The orientation is an initial condition riding an anchor (V); the anchor-supplier, if any, lives outside the audited symplectic jurisdiction (REG-13); the settling mechanism’s non-Bogoliubov character is proven in scoped form with one road unclosed (REG-14); the e_R boundary’s best candidate resolution (the second wall) is a question about the scale line’s state-governed profile; and the measurement stratum long ago placed the relational, gravity-flavored character of the framework in the folium. Program ω asks, in frozen order: **what does the corpus already assert about the state (A); does the scale line carry a second wall (B); can a settling dynamics be concretely constructed and does it move depth (C); and does anything in the state stratum supply the anchor, or is the initial-condition verdict terminal (D)?**

Scope limits, fixed now. No spacetime claim anywhere; the folium’s gravitational character may be *observed* where it appears but is not developed; Reading B stands; Track P untouched; generation structure enters only if a stage forces it. Roles unchanged: TOM builds, Fable-5 reviews and independently reimplements; either side’s hand-derivations carry their flags.

1. Stage A — The ω -dossier (documentary; the state’s own checklist)

Precedent: the $\mathbb{O}e_8$ necessity audit, which found C1 on its first pass. Both sides **independently** compile everything the corpus asserts about states — the quasi-free family and χ (Foundation), the settling ledger and its scoped theorem (Foundation §[LEDGER], V §4), thermality/KMS and the derived flow’s stationarity, the folium/events/relational structure (measurement paper), the vacua manifold and mixed endpoint (V §3), the carrier verdict and the anchor (V §5), the wall profile $m(u)$ ’s provenance (II/III), and the type-III citations — then **diff the dossiers**. Deliverable: one reference document with every claim

tagged and cited; every contradiction or gap between strata written months apart is a finding. Completeness criterion: review target 3.

2. Stage B — The second wall (the program’s agreed first physics assignment; the kinematic/dynamic split itself first proposed in `memory_note_second_wall.md`, filed before this plan existed and uncited in it until now — same split, same reasoning, no result asserted there, so B1 below is genuinely open, not a rerun)

B1 (kinematic reconnaissance; bounded; no ω -machinery). Against the closed record only: characterize the scale line’s two asymptotics (the $\mu \rightarrow 0$ contraction end; the opposite end as the record has it) and determine whether the structures already derived **force, admit, or exclude** a second zero-crossing of the mass profile. Pre-priced outcomes: *forced* (the second wall exists structurally — the `e_R` boundary converts to geography; the mirror-wall content theorem becomes B1’s immediate successor); *excluded* (a no-go worth its own note; `e_R` joins the named boundaries); *undetermined-without-dynamics* (the honest expected case per the reviewer’s separability warning — B2 activates inside Stage C, not before).

B2 (dynamic; gated on C). The profile question asked of an actual settling candidate: does the constructed dynamics generate a profile with one crossing or two? Not scheduled until C supplies a candidate.

Standing payoff, gated and named: if the second wall lands, Yukawa-as-overlap enters the registry as a derived-mechanism candidate (L-wall/R-wall overlap through the state-governed bulk). Nothing in this plan asserts it.

3. Stage C — Constructing a settling candidate (the program’s core)

Framing, fixed in advance to keep the corpus honest: Becoming’s ledger states no settling mechanism is currently supplied; V proved the specified class is a carrier. Stage C therefore performs **candidate-construction, not derivation** — building a concrete dynamics from the framework’s own materials and testing it, with every result tagged as a property of the candidate, never of the framework, unless uniqueness is separately proven.

C1 — the preservation lemma (REG-14), a genuine proof attempt. Unitaries mapping the quasi-free family into itself are Bogoliubov: attempt the proof (or locate and verify it in the literature per the cite-and-verify standard). Success upgrades Foundation’s trichotomy from quotation to theorem and closes the fourth road; failure documented is a finding.

C2 — the natural candidate: dissipative dynamics with zero-divisor jump operators. The framework supplies exactly one non-invertibility structure, and V measured it precisely: the anchored tunnel. C2 builds the Lindblad-

type semigroup with jump operator L_A (anchor A on the locus), on the finite-mode system first, and asks the pre-registered questions: **(i)** does it move χ (depth-motion — the defining settling property; the dynamics is non-unitary, so the scoped theorem permits it; whether it *does* is the test); **(ii)** is its motion directed (an H-theorem/monotonicity check — settling should settle, not wander); **(iii)** what does it do to the wall sector’s vacua manifold (the carrier verdict at dynamical level: pre-registered — the anchored dynamics transports the anchor’s fiber-sign into the state’s orientation datum, the dial turning *because history set it*, never spontaneously); **(iv)** the scalar-avoidance corollary at dynamical level (the kernel never touches the real units — does the dissipation therefore conserve the lone-scalar channel? A sharp, checkable question with measurement-stratum resonance). Instruments: the χ -meter (symplectic spectrum; exists from Stage C of Phase 5), the vacua-manifold reader (V §3), both re-registered under P6 with their extensions to non-unitary dynamics explicitly checked — the Stage-C addendum of V applies with full force here.

C3 — the family question. If C2’s candidate settles: is it distinguished within its class (anchor-averaged version, per V, is orientation-blind — is it also χ -static? Pre-registered by the $T_{avg} = 2 \cdot Id$ result: the democratic semigroup’s motion, if any, is pure type-dynamics), and what selects the anchor remains exactly the open question — C3 documents what the candidate *cannot* decide, in V’s carrier vocabulary.

4. Stage D — The anchor-supplier question in its own territory (gated on A-C; exploratory, strictest honesty clauses)

With the dossier, the wall verdict, and a working candidate: search the non-symplectic stratum — the only unaudited territory — for anything that supplies the anchor. Pre-priced outcomes: *a supplier with derived provenance* (the corpus’s largest possible finding; extraordinary claims protocol applies — everything two-implemented, the Phase 4 provenance audit re-run on the new object before any announcement); *no supplier, with the search-space characterized* (the boundary theorem in strengthened form: the initial-condition verdict extended into the state stratum — pre-registered as the *expected* outcome, so that finding it is confirmation, not disappointment); *ill-posedness of the search itself* (documented, returned to the registry). Stage D cannot run before C’s instruments exist and its honesty framing is proven in use.

5. Outcomes, pre-priced at program level

O1 — dossier + wall verdict + a settling candidate that moves χ with the carrier property confirmed dynamically: the state stratum acquires its first concrete dynamics, and the corpus’s story becomes end-to-end executable. **O2** — the candidate fails to settle: the mechanism class narrows again; the failure characterized is the finding (the pattern of V honored). **O3** — B1 excludes the

second wall: a named no-go; the e_R boundary becomes permanent structure. **O4** — D stays gated or lands on the boundary theorem: the program closes honestly on A-C, with the initial-condition verdict deepened rather than escaped. Every outcome produces a document.

6. Post-review amendments (round 1 — all five points accepted)

P1 (C2's branch choice, justified rather than silent). Foundation's trichotomy collapses, on inspection, into **two constructive branches**: the dissipative and open-system branches are one class (a Lindblad semigroup is an open system's reduced dynamics, by dilation), so the real fork is **Lindblad versus restriction**. The reviewer's point stands: a restriction map narrowing support over the scale line is arguably the more *literal* reading of "settling" — and the corpus even supplies its natural form (conditional expectation onto scale-truncated subalgebras, riding the half-sided structure). The order is therefore justified by **readiness, not naturalness**: the dissipative candidate's ingredients are already built, two-implemented, and orientation-characterized (V 's L_A , T_A , P_{kernel}), and its instruments extend checkably (P3); the restriction candidate requires new objects. C2 runs dissipative-first; **C2' names the restriction candidate as the deferred second, not a dropped alternative**, activated if C2's candidate fails to settle or as the comparison once it succeeds. Depth-motion (χ) remains the *defined* settling signature per Foundation regardless of branch.

P2 (B1 given decidable test shapes — the parity criterion). "Forced" and "excluded" now have exact forms. B1's actual task: **determine, from the closed record alone, the sign of the mass structure at each of the scale line's two asymptotics**. Then parity decides: *same sign at both ends* + the one established crossing $\Rightarrow$ an even crossing count $\Rightarrow$ a second wall **forced**; *opposite signs* $\Rightarrow$ odd count, minimal one, with any structural monotonicity argument then **excluding** a second; *sign fixed at only one end or neither* $\Rightarrow$ **undetermined**, honestly, with the specific missing datum named. The stage is now a two-sign determination with a parity theorem behind it — either the record contains the two signs or it doesn't, and B1's report states which.

P3 (the Gaussianity gate — Stage C's precondition, per the reviewer's hold). The χ -meter is well-defined along the candidate's flow only if the dynamics is Gaussian-preserving, which requires **jump operators linear in the mode operators**. The implementation is pinned accordingly: the anchor's dissipation enters as the set of linear jump operators built from L_A 's matrix elements ($c_j = \sum_k (L_A)_{jk} a_k$), not as any quadratic one-body form. Under this choice the covariance matrix obeys the standard closed-form ODE and $\chi(t)$ is well-defined (though no longer conserved — which is the point). **C2's first computation, before any settling question is asked**: verify Gaussianity-preservation and the covariance ODE on the finite-mode system, and re-register the χ -meter and vacua reader under P6 with this extension

explicit. No χ -claim is admissible before this gate passes.

P4 (Stage D's effort clause, the reviewer's wording adopted). "Expected" governs how the null finding is *reported*, never how hard it is *looked for*: the search-space characterization accompanying a null result must meet the same completeness bar it would need if a positive result were expected.

P5 (the extraordinary-claims protocol, defined here since it gates the program's largest possible finding). Any candidate anchor-supplier triggers, in order: (i) full two-implementation of the object and its entire provenance chain; (ii) the Phase 4 sign-provenance audit re-run on the new object (derived versus convention versus gauge, by the standing instruments); (iii) pre-registration of the claim's falsifiers before any confirming computation is accepted as such; (iv) a dedicated hostile-review round on the claim alone, separate from any stage review; (v) entry into the corpus at [CANDIDATE] weight for one full review cycle before any upgrade. Precedents: the Phase 4 audit vocabulary and V's carrier-verdict process, both of which this protocol generalizes.

Freeze protocol: §§0-6 to the reviewer; the reviewer's hold was specifically on P2's criteria and P3's gate — both now concrete; confirmation on sight freezes, with Stage A (the dossier compilations) as the frozen program's first logged work.